Annasaheb Dange College of Engineering and Technology, Ashta
(An Autonomous Institute affiliated to Shivaji University, Kolhapur)
Department of Computer Science and Engineering
(NAAC A++ Grade Accredited Institute, NBA Accredited Program, ISO 9001: 2015 Certified Institute)

Micro-Project Report

2CSVS308 : Python Programming

Canteen Billing System And Analysis Using Python

Submitted By :

1) Shrutika Anilkumar Yadav (1024031131)

2) Vrushali Gopichand Wadar (1024051091)

T.Y. B.Tech in Computer Science and Engineering

Under the Guidance of : Prof. Shrihari D. Khatavkar

Department of Computer Science and Engineering

Annasaheb Dange College of Engineering and Technology, Ashta 416 301

Project Title : Canteen Billing And Analysis Using Python

1. Introduction

Food wastage is a common problem in college canteens due to uncertain food demand. Sometimes more food is prepared than required, which leads to wastage and loss. **Canteen Billing And Analysis** r is a simple project made in Python that helps to solve this problem.

In this project, the user enters food item details, quantity sold, selling price and cost. The program uses previous sales records to estimate the expected demand for different food items. It also keeps records of daily sales so that the canteen can check its previous performance.

The project also includes a finance section that calculates total sales, total cost and profit or loss. Pandas and NumPy are used for data analysis and numerical calculations. This project helps reduce food wastage, manage food demand and understand the financial performance of the canteen.

2. Objectives

The main objectives of the project are:

1. To develop a simple food demand prediction system using Python.
2. To estimate the required quantity of food using previous sales records.
3. To maintain records of food items, quantities, prices and sales.
4. To calculate total sales, cost and profit or loss.

2.1 Objective 1: To Develop a Simple Canteen Billing And Analysis Using Python.

Implementation: The system uses the previous sales data of a food item to predict its future demand. It calculates the overall average demand and the average demand of the most recent three sales. It gives 60% importance to the overall average and 40% importance to the recent demand to calculate the predicted demand, and then adds 10% safety stock to give the recommended quantity to the user.

Procedure:

1. The system gets the previous sales quantities of the food item from the database.
2. The overall average demand is calculated from the sales data.
3. The average demand of the most recent three sales is calculated.
4. The predicted demand is calculated by giving 60% importance to the overall average and 40% importance to the recent demand.
5. A 10% safety stock is added and the recommended quantity is given to the user.

Technology Used: Python – for implementing the demand prediction logic; NumPy – for calculating averages and numerical values; SQLite – for storing the previous sales data.

Result: The system predicts the future demand of a food item from its previous sales. For example, if the system predicts that approximately 50 units will be required, it recommends a slightly higher quantity so that the canteen can handle a small increase in demand.

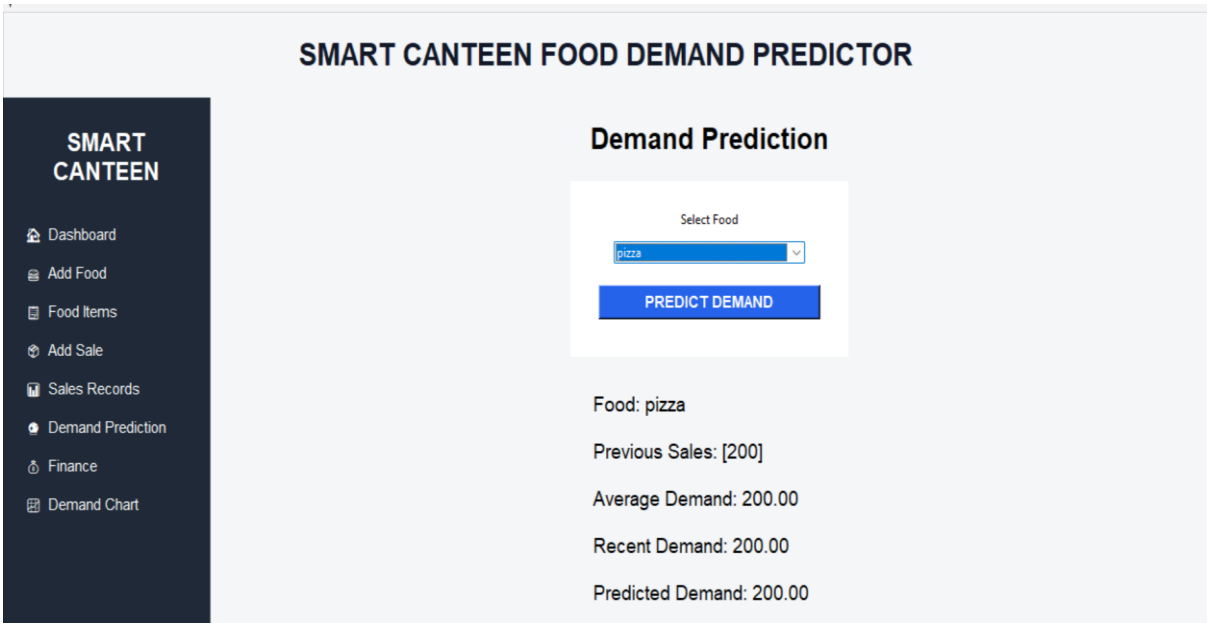


Fig 1. Food Demand Prediction Output

2.2 Objective 2: To Estimate the Required Quantity of Food Using Previous Sales Records.

Implementation: The system estimates how much food should be prepared using the previous sales records stored in the database. It studies the previous sales and calculates the average demand. It also considers the recent three sales because recent sales give a better idea of the current demand. After calculating the predicted demand, the system adds 10% safety stock and gives the final value as the recommended quantity.

Procedure:

1. The system reads the previous sales records of the food item from the database.
2. The average demand is calculated from the previous sales.
3. The most recent three sales are considered to understand the current demand.
4. The predicted demand is calculated from these values.
5. A 10% safety stock is added to the predicted demand to get the recommended quantity.

Technology Used: Python – for implementing the estimation logic; NumPy – for calculating the average demand and quantities; SQLite – for storing and reading the sales records.

Result: The canteen gets an estimated quantity that it can prepare based on previous demand instead of preparing food randomly. For example, if the predicted demand is 50 units, the system adds 10% extra and recommends approximately 55 units.

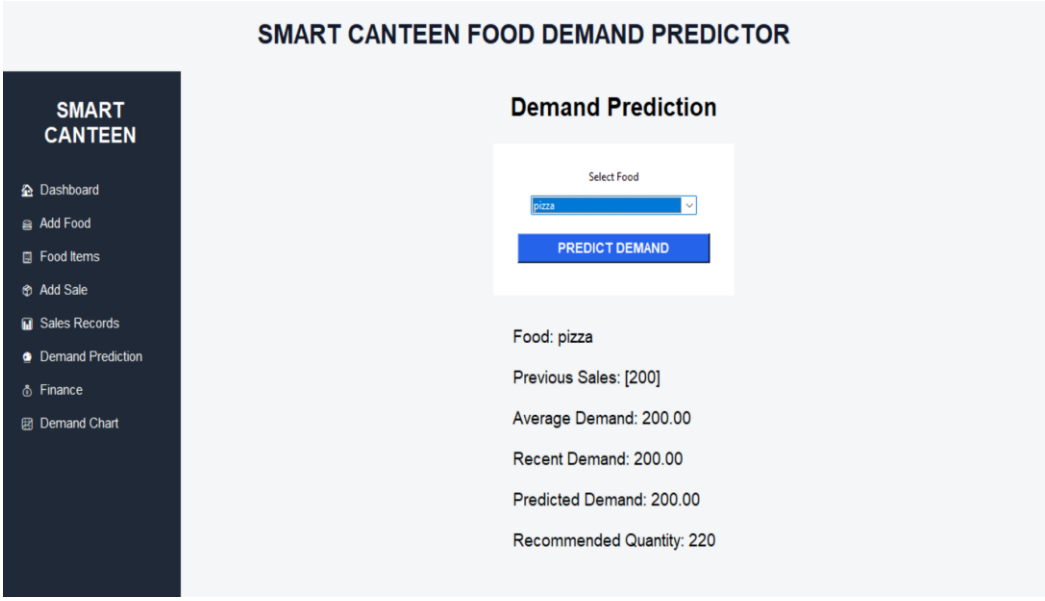


Fig 2. Recommended Quantity Output

2.3 Objective 3: To Maintain Records of Food Items, Quantities, Prices and Sales.

Implementation: The system uses an SQLite database to maintain proper records of food items and sales. A food items table stores the food name, selling price and cost price, and a sales table stores the food ID, sale date and quantity sold. The user can add a new food item, view food items, add a sale, view sales and delete a food item through the system.

Procedure:

1. The user enters the food name, selling price and cost price to add a new food item.
2. The system stores the food item details in the food items table of the SQLite database.
3. When a sale is added, the system stores the food ID, the quantity sold and the current date in the sales table.
4. The user can view the stored food items and sales records.
5. The user can delete a food item through the system when it is no longer required.

Technology Used: Python – for implementing the record management functions; SQLite – for storing food items and sales records; datetime – for storing the sale date.

Result: The system keeps proper records of food items, prices and daily sales. These stored records are important because the same sales data is later used for demand prediction and financial analysis.

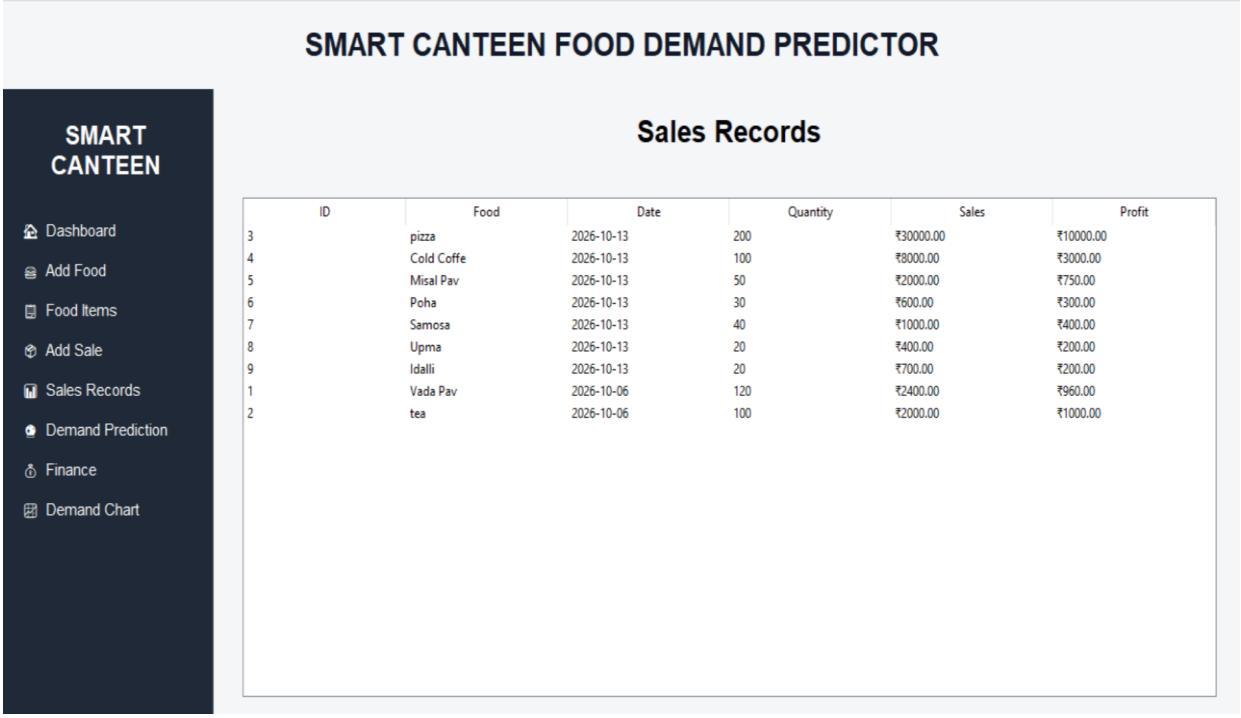


Fig 3. Food Items and Sales Records

2.4 Objective 4: To Calculate Total Sales, Cost and Profit or Loss.

Implementation: The system calculates the financial performance of the canteen. Pandas is used to read and organize the sales data. Total sales are calculated by multiplying the quantity sold by the selling price, total cost is calculated by multiplying the quantity sold by the cost price, and profit is calculated by subtracting total cost from total sales.

Procedure:

1. The sales data is read from the database and organized using Pandas.
2. Total sales are calculated by multiplying the quantity sold by the selling price.
3. Total cost is calculated by multiplying the quantity sold by the cost price.
4. Profit is calculated by subtracting the total cost from the total sales.
5. The total sales, total cost, total profit and profit margin are calculated for all the recorded sales and displayed.

Technology Used: Python – for implementing the finance calculations; Pandas – for reading, organizing and analyzing the sales data; SQLite – for storing the sales records.

Result: The finance report shows the total sales, total cost, total profit and profit margin, so the canteen can understand how much it earned, how much it spent and how much profit it made.

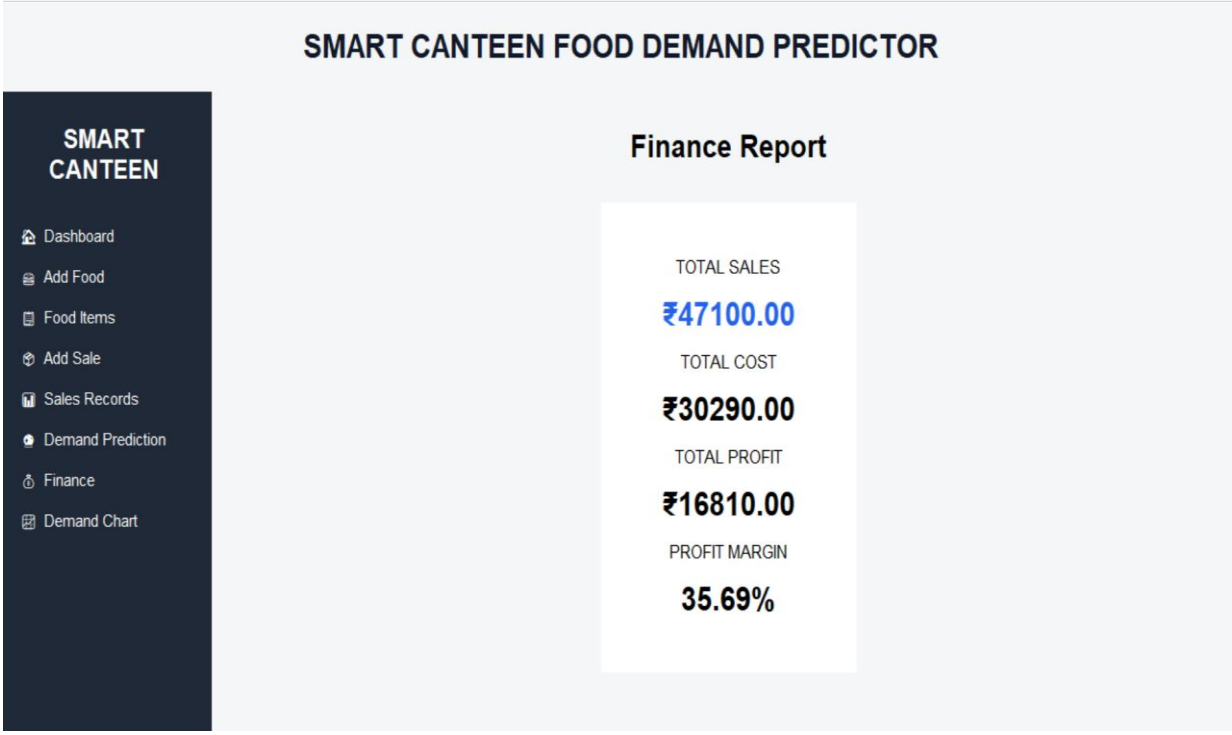


Fig 4. Finance Report Output

3. Methodology

The project is developed step by step, as explained below:

1. **Input Collection:** The user enters the food item, price, cost and quantity sold.

2. **Data Validation:** The program checks the entered information and handles incorrect input.

3. **Demand Calculation:** Previous sales records are used to calculate average demand and estimate the required quantity of food.

4. **Finance Calculation:** The system calculates total sales, total cost and profit or loss.

5. **Data Storage:** Daily food and sales records are stored using file handling.

6. **Data Analysis:** Pandas is used to organize and analyze the stored sales records.

7. **Numerical Calculation:** NumPy is used for calculations such as average demand and quantities.

8. **Output:** The expected demand, sales details, cost, profit or loss and other information are displayed to the user.

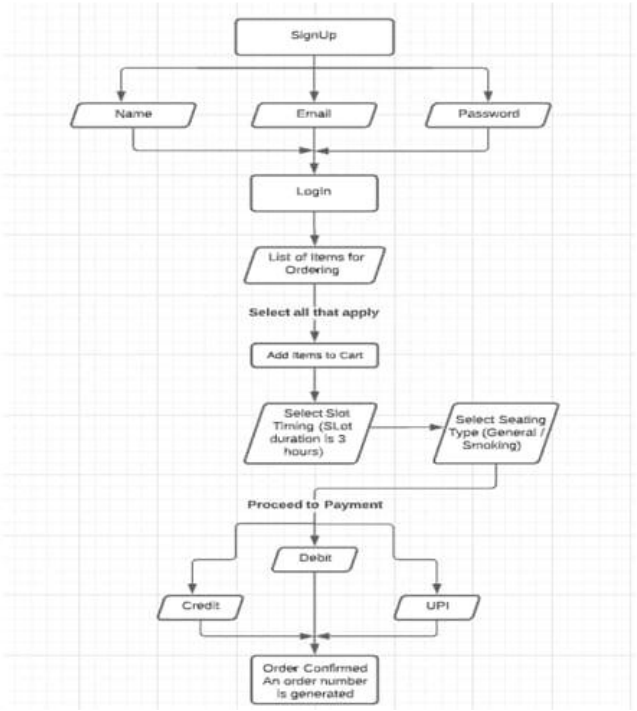


Fig 5. Workflow Diagram

4. Technology Used

Technology	Purpose
Python	Main programming language
Pandas	Data analysis and finance report calculations
NumPy	Numerical calculations such as average demand
SQLite	Store food items and sales records
File Handling	Store daily food and sales records
datetime	Manage sale dates and current date
Python Concepts	Functions, Loops, Conditions, OOP and Exception Handling

5. References

- Python Documentation – Python Software Foundation – docs.python.org
- NumPy Documentation – numpy.org
- Pandas Documentation – pandas.pydata.org
- GeeksforGeeks – Python Programming and Data Analysis – www.geeksforgeeks.org

6. Conclusion

The Canteen Billing And Analysis provides a simple and useful solution for managing food demand and sales in a college canteen. The system uses previous sales records to estimate future food demand and recommends the required quantity with a small safety stock. This can help the canteen avoid preparing excessive food and reduce food wastage.

The system also maintains food and sales records using an SQLite database. It stores important information such as food items, selling price, cost price, sale date and quantity sold. The stored data is further used for demand prediction and financial analysis.

Pandas is used to organize and analyze the sales data, while NumPy is used for numerical calculations. The system also calculates total sales, total cost, total profit and profit margin, helping the canteen understand its financial performance.

Overall, the project demonstrates how Python programming concepts can be used to develop a practical canteen management solution that supports demand estimation, sales record management and financial analysis.